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Improving Efficiency with Orthogonal Exploration for Online Robotic Assembly Parameter Optimization

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***Abstract*— In this paper, an online robotic assembly parameter optimization method is developed, this method make industrial robots can assemble workpiece from unskilled to skilled just like human's learning behavior. With the development of the force sensor technology and force control technology, the industrial robots are used in high precision assembly tasks which are more complicated. In order to ensure the efficiency and success rate of assembly, it is necessary to select the appropriate assembly parameters, this problem is called robotic assembly parameters optimization. The traditional solutions are used by artificial methods, a lot of experiments are carried out to get the optimal parameters, which are very time-consuming and laborious. Especially when the production line changes, using the traditional solutions have to do heavy experiments again, it can't meet the requirements of today's flexible manufacturing requirements. This paper presents an online robotic assembly parameter optimization method, which is called Gaussian Process Regression surrogated Bayesian Optimization Algorithm based on the Orthogonal Exploration (OE-GPRBOA), this method can liberate the labor, does not require artificial participation. The algorithm can optimize the parameters autonomously, finally find the optimal parameters for robotic assembly. For GPR is suitable for processing high dimension, small size of sample and nonlinear complex regression problems, the proposed OE-GPRBOA method can be used for various assembly tasks. In this paper, peg-in-hole assembly experiments are performed. The proposed method also compared with design of experiments (DOE) method and GPRBOA method. Experimental results show that, the proposed OE-GPRBOA method has more efficiency to find the optimal assembly parameters, this method can generate big economic impact.**

I. INTRODUCTION

A. Background

Although industrial robots have been widely used in the field of assembly, it is not competent for high precision assembly tasks based on position control. With the development of force sensor technology and force control technology, industrial robots are gradually applied to high precision assembly tasks, but it still faces great challenges. For example, installing an valve into a valve body is not always as easy as it looks. As show in the Fig.1, the radius of the valve is about 24.96 mm while the radius of the hole in the valve body is 25.00 mm with a tolerance about 40μm. Because of the fixture errors, the valve cannot be aligned with the holes on the valve body exactly, it must adjust the position and posture of valve, the adjustments are very tiny, if any little inappropriate adjustment, will lead to the failure of the whole assembly, the valve will be stuck at the surface of the valve body. To assemble the valve successfully, several parameters are involved, such as search force, search speed, search radius, the insertion force. If we want the industrial robot assemble workpiece successfully in the shortest time, assembly parameters play a crucial role. Therefore, to improve the success rate and efficiency of the assembly, it is important to search the optimal assembly parameters, this process is called robotic assembly parameters optimization.

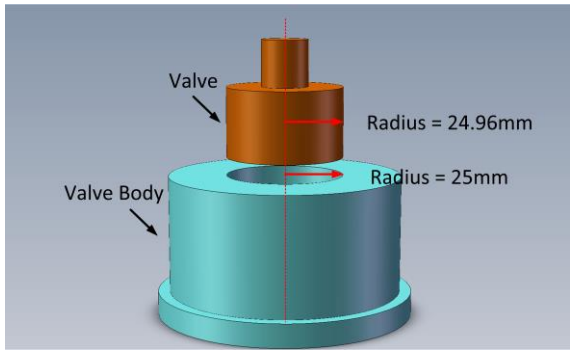


Figure 1. The high precision valve body assembly process.

B. Research Status

There are two main classes of the assembly parameters optimization methods currently, one are the offline methods[1][2], the other are online methods. The offline methods includes two kinds methods which are design-of-experiment (DOE)[3][4] and genetic algorithm (GA)[5][6]. DOE is a systematic method of parameter optimization. Through the design of a series of experiments, the optimal assembly parameters are selected. Genetic algorithm is a randomly search for optimal parameters, in order to improve the search efficiency and reduce non-essential parameter space search, Artificial Neural Network (ANN) is utilized to model whether the parameters are "good" or "bad" to filter the candidate parameters first without performing any experiment[7][8]. Offline methods' advantage is the optimization result has high reliability because of a lot of experiments, but off-line methods' disadvantage is also very obvious that are heavy task of experiment, low efficiency, when the assembly line operation change, the heavy task of experiment should do again, very time-consuming. Moreover, at off-line design process, production lines are all stop work, cause wasting resources of facilities.

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Therefore, the on-line optimization method is put forward, which can autonomous search for the optimal assembly parameters online and make the robot complete the parameter optimization process online independently, thus the optimization process don't need artificial participation. At present, the method of research is still in its infancy, H. Cheng and H. Chen first used the Gaussian process regression surrogated Bayesian optimization algorithm (GPRBOA) to solve the online parameter optimization, and has preliminary achieved the goal of autonomous optimization online for multi-parameters[9], but the proposed method is still not perfect, such as low efficiency, low success rate, low robustness, cannot solve the multi-objective optimization problem, etc.

C. Summary

According to the orthogonal experimental design thought, we unified the Orthogonal Exploration(OE), Gaussian Process Regression (GPR) and Bayesian Optimization Algorithm (BOA) , and proposed the robot online assembly parameters optimization method called OE-GPRBOA, to solve low efficiency, low success rate and low robustness problems of GPRBOA method. The method is verified by the experimental results, and the results show that the method can greatly improve the efficiency and success rate to get optimal assembly parameters.

II. Problem Definition

The Fig.2 show a typical robotic system to perform a high precision peg-in-hole assembly process. A searching method is used to find the exact location of the workpiece. After the part is engaged with the workpiece , an insertion force is applied to insert the part into the workpiece . During insertion, the tool orientation is changed according to the orientation of the workpiece to avoid sticking. The assembly parameters involved include search force, search speed, search radius, the insertion force.

III. OE-GPRBOA

A. Core idea

The goal of the optimization of the assembly parameters is to find a suitable assembly parameters combination, so that the robot can complete the assembly task in the shortest time. As show in Fig.3 We can take the assembly process as an unknown mathematical model $\mathbf{Y}=f(\mathbf{X})$, where X is the multi-dimensional input vectors, for various assembly parameters which would affect the performance of the assembly, $\mathbf{Y}$ is the output variable, for the cycle time of assembly. Then the assembly parameters optimization

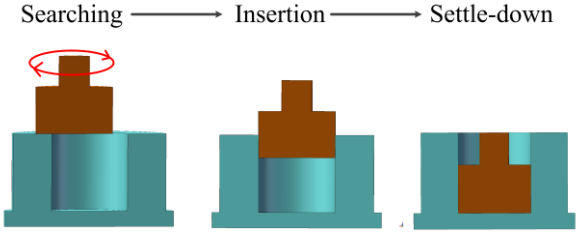


Figure 2. High precision peg-in-hole assembly process.

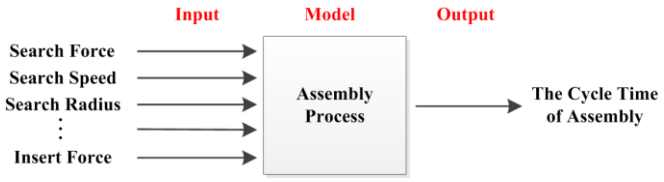


Figure 3. High precision peg-in-hole assembly process.

problem can be described as: how to find the appropriate $\mathbf{X}$ as soon as possible to get the minimum value of $\mathbf{Y}$. In this paper, the solving method of the problem: by selecting the suitable training data $\mathbf{X}$, using a machine learning method to learn the unknown model $\mathbf{Y}=f(\mathbf{X})$, then for the learned model $\mathbf{Y}=f(\mathbf{X})$, the optimization algorithm is used to obtain the optimal solution $\mathbf{X}$. In this paper, the selection method of training data are based on orthogonal exploration, and the method of modeling is Gauss process regression, the optimization method uses the Bayesian optimization algorithm.

B. Gaussian process regression

Gaussian process regression (GPR)[10] is a new machine learning method based on Bayesian theory and statistical learning theory, is a nonparametric tool that can deal with modeling problems with noisy observation and system uncertainties, suitable for processing high dimension, small size of sample and nonlinear complex regression problems, have good adaptability and strong generalization ability. It has become a hot research field of machine learning in the world, and has been successfully applied in many fields.

From function-space view, Gaussian Process(GP) is a set of arbitrary finite random variables with a combined Gaussian distribution, which is completely determined by mean function and covariance function.

$$\begin{cases} m(\mathbf{x})=E[f(\mathbf{x})] \\ k(\mathbf{x},\mathbf{x}')=E[(f(\mathbf{x})-m(\mathbf{x}))(f(\mathbf{x}')-m(\mathbf{x}'))] \end{cases} \tag{1}$$

$\mathbf{x},\mathbf{x}'\in R^d$ are arbitrary random two variables. GP can be defined as $f(\mathbf{x})\sim \mathrm{GP}(m(\mathbf{x}),k(\mathbf{x},\mathbf{x}'))$. In order to have a simple symbol, the data are usually preprocessed, and make the mean function equal to 0.

For regression problems, consider the model: $\mathbf{y}=f(\mathbf{x})+\varepsilon$, where $\mathbf{x}$ is the input vector, f is a function value, $\mathbf{y}$ is the observation value with noise pollution, and further assume the noise as $\varepsilon\sim N(0,\sigma_n^2)$. We can obtain the prior distribution of observed values is $\mathbf{y}\sim N(0,K(\mathbf{X},\mathbf{X})+\sigma_n^2 I_n)$, as well as the joint prior distribution of the observed and predicted data are

$$\begin{bmatrix}\mathbf{y}\\ \mathbf{f}_*\end{bmatrix}\sim N\left(0,\begin{bmatrix}K(\mathbf{X},\mathbf{X})+\sigma_n^2 I_n & K(\mathbf{X},\mathbf{x}_*)\\ K(\mathbf{x}_*,\mathbf{X}) & k(\mathbf{x}_*,\mathbf{x}_*)\end{bmatrix}\right) \tag{2}$$

Where $K(\mathbf{X},\mathbf{X})=K_n=(k_{ij})$ is a $n\times n$ symmetric and positive definite covariance matrix, the matrix element $k_{ij}=k(\mathbf{x}_i,\mathbf{x}_j)$ is used to measure the correlation between $\mathbf{x}_i$

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and $\mathbf{x}_j$; $K(\mathbf{X},\mathbf{x}_*)=K(\mathbf{x}_*,\mathbf{X})^T$ is the $n\times 1$ covariance matrix between the test point $\mathbf{x}_*$ and the input of training set $\mathbf{X}$; $k(\mathbf{x}_*,\mathbf{x}_*)$ is self-covariance for the test point $\mathbf{x}_*$; I_n is n dimensions unit matrix.

The posterior distribution of the predicted values $\mathbf{f}_*$ can be calculated.

$$\mathbf{f}_* \mid \mathbf{X},\mathbf{y},\mathbf{x}_* \sim N\left(\bar{\mathbf{f}}_*, \operatorname{cov}(\mathbf{f}_*)\right) \tag{3}$$

where

$$\bar{\mathbf{f}}_* = K(\mathbf{x}_*,\mathbf{X})[K(\mathbf{X},\mathbf{X})+\sigma_n^2 I_n]^{-1}\mathbf{y} \tag{4}$$

$$\operatorname{cov}(\mathbf{f}_*) = k(\mathbf{x}_*,\mathbf{x}_*) - K(\mathbf{x}_*,\mathbf{X})\times[K(\mathbf{X},\mathbf{X})+\sigma_n^2 I_n]^{-1}K(\mathbf{X},\mathbf{x}_*) \tag{5}$$

Then $\hat{\mu}_* = \bar{\mathbf{f}}_*$ and $\bar{\sigma}_{f_*}^2 = \operatorname{cov}(\mathbf{f}_*)$ are the mean and variance of the predicted value $\mathbf{f}_*$ corresponding to the test points $\mathbf{x}_*$.

GPR can choose different covariance functions, the common covariance function is the square index covariance, which is

$$k(\mathbf{x},\mathbf{x}')=\sigma_f^2 \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{x}')^T M^{-1}(\mathbf{x}-\mathbf{x}')\right) \tag{6}$$

where $M=\operatorname{diag}(l^2)$, l is a characteristic length-scale, σ_f^2 is the variance of signal, the parameter set $\theta=\{M,\sigma_f^2,\sigma_n^2\}$ is the hyperparameters, which is obtained by the maximum likelihood method. First of all, establishing a negative log likelihood function of the conditional probability of the training samples $L(\theta)=-\log p(\mathbf{y}\mid\mathbf{X},\theta)$, and solve the partial derivative of $L(\theta)$. Then acquire the optimal solution of the hyperparameters by minimizing the partial derivatives, this optimization problem can be solved using different techniques such as Conjuncted Gradient Algorithm or evolutionary algorithms. Here, the partial derivative of negative log likelihood function $L(\theta)$ and the hyperparameters θ are as follows:

$$L(\theta)=\frac{1}{2}\mathbf{y}^T C^{-1}\mathbf{y}+\frac{1}{2}\log|C|+\frac{n}{2}\log 2\pi \tag{7}$$

$$\frac{\partial L(\theta)}{\partial \theta_i}=\frac{1}{2}\operatorname{tr}\left((\boldsymbol{\alpha}\boldsymbol{\alpha}^T-C^{-1})\frac{\partial C}{\partial \theta_i}\right) \tag{8}$$

Where $C=K_n+\sigma_n^2 I_n$, $\alpha=(K+\sigma_n^2 I_n)^{-1}\mathbf{y}=C^{-1}\mathbf{y}$, After acquire the hyperparameters, we can get predicted value $\mathbf{f}_*$ and its variance $\bar{\sigma}_{f_*}^2$ corresponding to the test points $\mathbf{x}_*$ by using the (4) and (5).

C. Bayesian Optimization Algorithm

The Bayesian Optimization Algorithm(BOA)[11] estimates a probability distribution of promising solutions using Bayesian Network(BN) in order to generate new candidates. In this paper, BN model is replaced by the GPR model for decrease the computational complexity and difficulty.

In the GPR surrogated BOA (GPRBOA)[12][13] can iteratively model a complex system and optimize the system performance. In each iteration, new samples are added into the existing data set and used to update the GP model. The new GP model will then be used to search for the optimal solutions by maximizing a performance index s(x)(also known as the acquisition function) over the input domain. Because the acquisition function controls the new sample points, it directly affects the quality of the built model and the optimal solution. Therefore, it plays a key role in the GPRBOA algorithm. This paper use the Lower Confidence Bound(LCB)[14] as acquisition function to acquire a new candidate. LCB is recently developed based on the idea of exploiting lower confidence bounds (upper, when considering maximization) to increase the optimization efficiency. It has the form

$$LCB(\mathbf{x}) \triangleq \mu(\mathbf{x})-\kappa\cdot\sigma(\mathbf{x}) \tag{9}$$

where κ is a scaling factor. Instead of only sampling the points with minimum mean $\mu(\mathbf{x})$ or maximum variance $\sigma(\mathbf{x})$ predicted using the current model to improve the model uncertainty.

D. The Selection of X

To completely solve the problem of assembly parameter optimization, not only to solve the modeling problem and optimization problem, but also need to solve the following three problems: the global exploration problem of the parameter space, the local exploitation problem of the parameter space, the balance problem between exploration and exploitation.

1) Exploration

The global exploration of the parameter space: when we model the assembly process, we need a certain amount of training samples, how to select appropriate training samples in the global scope of the parameter space is called exploration problem, for this problem, our goal is to use as little as possible training samples to learn accurate and complete model, improve the efficiency of learning model.

For the global exploration of the parameter space, Cheng's method used random exploration, random exploration is a scattershot approach, resulting in parameter space exploration is not uniform, some parts of parameter space are explored too little, some parts of parameter space are explored too much, therefore, this exploration method is inefficient. In this paper, the orthogonal exploration method based on orthogonal experimental design is used for the exploration of the parameter space, the orthogonal experimental design is a design method of multi factors and multi levels, it pick out a part of the representative points from comprehensive experiments according to the orthogonality, these have representative points with "uniformly dispersed, symmetrical comparability " characteristics, orthogonal experimental design is main method of analysis factor design, is an efficient, fast, economical experimental design method. In this paper,

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the exploration is shown as follow.

$$\mathbf{x}^* = \begin{cases} \arg L_n(\mathbf{x}) & i \le n \\ \arg rand(\mathbf{x}) & i \ge n, avg\sigma \ge k_\sigma \end{cases} \quad (10)$$

The n is the number of orthogonal exploration. The L_n is orthogonal table, used to produced orthogonal exploration assembly parameters. The $avg\sigma$ is average model learning deviation for all predict values, used to judge the convergence degree of the learning assembly model.

2) Exploitation

Second problem is the local exploitation problem of the parameter space: when the model were learned to a certain degree, the optimal parameter may be found in some local parameters space, how to further optimize in the local parameter space to find the optimal parameters, namely local exploitation problem of the parameter space.

To solve the local exploitation problem of the parameter space, we can get the **X** value which is the minimum value of $LCB(\mathbf{x})$ as the next time input of the assembly model.

$$\mathbf{x}^* = \arg\min LCB(\mathbf{x}) \quad i \ge n, avg\sigma \le k_\sigma \quad (11)$$

3) Balance

The balance problem between exploration and exploitation: when to explore the parameter space, when to exploit the parameter space, what is the switching condition between the exploration and exploitation, namely the balance problem between exploration and exploitation. In this paper, we use average model learning deviation $avg\sigma$ to judge the convergence degree of the learned assembly model. Another convergence criterion is the normalized change of hyperparameters $\Delta\theta = |\theta_k - \theta_{k-1}| / |\theta_k - \theta_0|$, which may be premature convergence of learned assembly model. Finally, the selection of new assembly parameter is shown as follows.

$$\mathbf{x}^* = \begin{cases} \arg L_n(\mathbf{x}) & i \le n \\ \arg rand(\mathbf{x}) & i \ge n, avg\sigma \ge k_\sigma \\ \arg\min LCB(\mathbf{x}) & i \ge n, avg\sigma \le k_\sigma \end{cases} \quad (12)$$

The algorithm of online robotic assembly parameter optimization is shown in Fig.4.

IV. EXPERIMENT

A. Experimental Setup

For GPR is suitable for processing high dimension, small size of sample and nonlinear complex regression problems, the online assembly parameters optimization method can be used for various assembly tasks. In order to verify the proposed method, we first use Cheng's peg-in-hole assembly experimental data[9] whose method is used by GPRBOA, and then compare the efficiency about DOE, GPRBOA and OE-GPRBOA, and finally get the conclusion. The peg-in-hole assembly process includes the spiral searching hole with guiding force, inserting with force control, settle-down. Three parameters involved: the search speed(Ss), search force (Fs) and insertion force (Fi), the values of 3 parameters and 3 levels are shown in the follow TABLE Ⅰ.

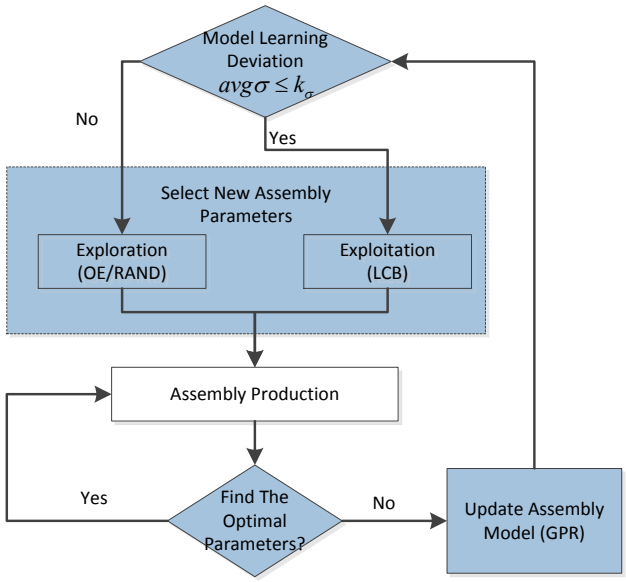


Figure 4. The OE-GPRBOA online robotic assembly parameter optimization algorithm.

TABLE I. THE VALUES OF 3 PARAMETERS AND 3 LEVELS

Level / Parameter	Ⅰ	Ⅱ	Ⅲ
Ss(mm/s)	250	300	350
Fs(N)	5	20	35
Fi(N)	50	75	100

B. Experimental Result.

1) Offline method

DOE method will do the experiment offline about various parameters combinations, and then select the optimal parameters, and finally use the optimal parameters in the assembly production. For the experimental design of 3 parameters and 3 levels, the totals parameters combinations are $3^3 = 27$. Because the cycle time is affected by several random factors, the performance of each set of parameters has to be statistically calculated. Therefore, each parameters combination should performed experiments 10 times, that would ensure the reliability and accuracy of final experimental results, therefore, the totals of experiments of 3 parameters and 3 levels are 270. The TABLE Ⅱ are the statistical results of the cycle time of robotic assembly under various combinations of parameters.

2) Online methods

Online optimization method which algorithm can optimize the parameters while robot assembling. When an assembly task is completed by robot, the related data will be update and used to learn assembly model and solve the optimal solution. After learning and optimization repeatedly, the optimal parameters will be acquired. This experiment compares two methods, one is GPRBOA based on the random exploration.

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The other is OE-GPRBOA based on orthogonal exploration.

TABLE II. DOE EXPERIMENTAL RESULTS[9]

Step	Ss	Fs	Fi	Tc(μ)	Tc(σ)	3σ	μ-3σ	μ+3σ
1	250	35	50	6	0.68	2.04	3.96	8.04
2	300	35	50	5.5	1.05	3.15	2.35	8.65
3	350	35	50	5.1	1.67	5.01	0.09	10.11
4	250	20	50	2.4	0.17	0.51	1.89	2.91
5	300	20	50	2.3	0.13	0.39	1.91	2.69
6	350	20	50	2.3	0.09	0.27	2.03	2.57
7	250	5	50	6.1	0.62	1.86	4.24	7.96
8	300	5	50	6.1	0.56	1.68	4.42	7.78
9	350	5	50	6	0.42	1.26	4.74	7.26
10	250	35	75	5.5	1.18	3.54	1.96	9.04
11	300	35	75	5.4	0.79	2.37	3.03	7.77
12	350	35	75	5.9	1.11	3.33	2.57	9.23
13	250	20	75	2.4	0.2	0.6	1.8	3
14	300	20	75	2.4	0.11	0.33	2.07	2.73
15	350	20	75	2.4	0.09	0.27	2.13	2.67
16	250	5	75	6.2	0.52	1.56	4.64	7.76
17	300	5	75	6	0.56	1.68	4.32	7.68
18	350	5	75	5.8	0.67	2.01	3.79	7.81
19	250	35	100	5.6	0.88	2.64	2.96	8.24
20	300	35	100	6.1	0.45	1.35	4.75	7.45
21	350	35	100	6.1	0.86	2.58	3.52	8.68
22	250	20	100	2.5	0.14	0.42	2.08	2.92
23	300	20	100	2.3	0.1	0.3	2	2.6
24	350	20	100	2.3	0.14	0.42	1.88	2.72
25	250	5	100	5.9	0.45	1.35	4.55	7.25
26	300	5	100	5.9	0.58	1.74	4.16	7.64
27	350	5	100	6	0.49	1.47	4.53	7.47

Two methods both can find the optimal parameters finally. As shown in the figure below, in the Fig.5, the left figure shows the model learning result after 27 times assembly, the right figure is real assembly model. In the Fig.6, the left figure show the change of normalized change of hyperparameters $\Delta\theta$, the $\Delta\theta$ is gradually reduce to zero, means that the assembly model is gradually converged. The right figure shows the change of average model learning deviation $avg\sigma$, is also gradually reduced, indicating that the assembly model is gradually converged and become stable. Fig.7 shows the change of cycle time of assembly every time, after 9th experiment, we find the optimal parameters, and the cycle time of assembly is shortest.

3) Result comparison

Although the GPRBOA and OE-GPRBOA both can obtain the optimal solution, but the different exploration methods and different model convergence criterion lead to different efficiency. GPRBOA is based on random exploration, and adopt the normalized change of hyperparameters $\Delta\theta$ as the learning model convergence criterion. OE-GPRBOA is based on orthogonal exploration, and adopt average model different methods, we do 100 groups experiments for GPRBOA and OE-GPRBOA respectively, every group the robot assemble workpiece 27 times. The TABLE Ⅲ is the number of experiments required to find the optimal parameters under two methods with the corresponding success rate. From the TABLEIII, we can see that OE-GPRBOA is more effective than DOE and GPRBOA, which can greatly improve the efficiency and success rate to find the optimal parameters.

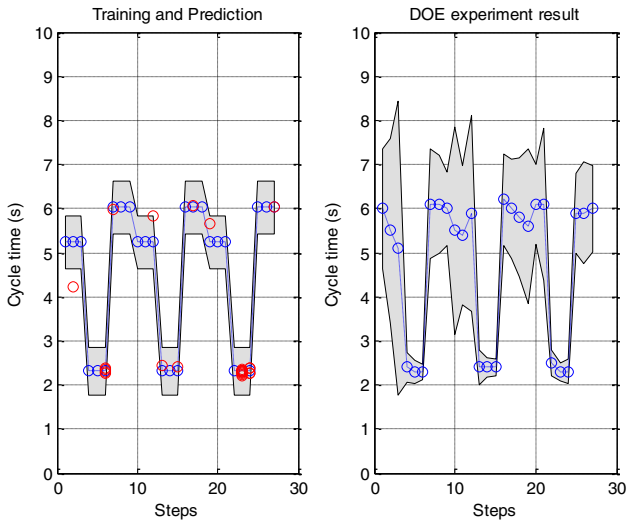


Figure 5. The learning and real model of assembly. (*the left is the learning model, and the right is the real model made by the* TABLE Ⅱ*. In the left figure, the red dots are samples data for the model learning, and the blue dot and grey areas are the model leaning results, the blue dots are mean u of the learning model and grey areas are the deviation range of the learning model.)*

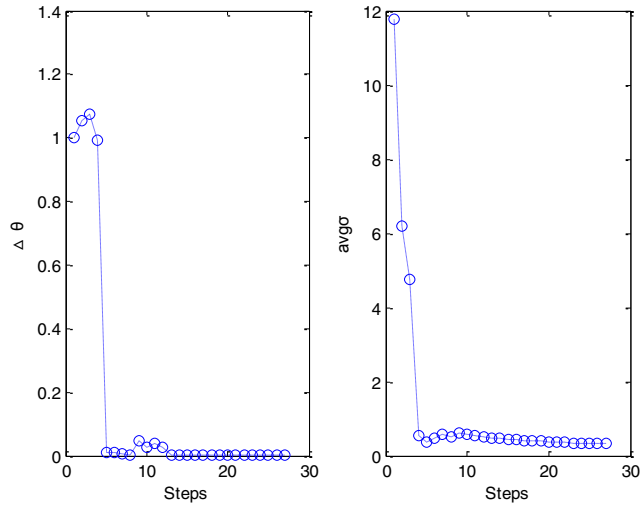


Figure 6. The normalized change of hyperparameters and the average model learning deviation.

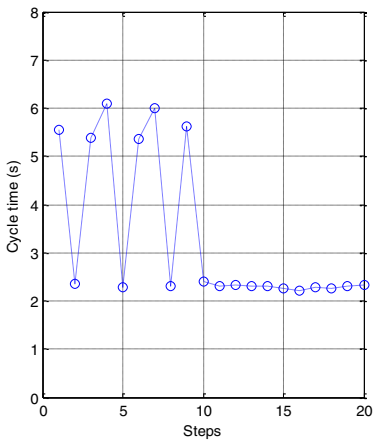


Figure 7. The cycle time of assembly.

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TABLE III. THE SUCCESS RATE OF DIFFERENT METHODS

Method	Assembly Times	Success rate
DOE	270	100%
GPRBOA	20	67%
	14	48%
	9	25%
OE-GPRBOA	9	99%

C. Discussion

The DOE method is a kind of comprehensive experimental method, experiments were conducted for every parameter combination, and in order to overcome the result uncertainty, the experiments of each parameter combination need to be conducted, the totals of experiments are $K\prod_{i=1}^{M} N_i$, M is the number of parameters, N is the levels of each parameter, K is the repeat times of each parameter combination, with the number of parameters and the levels of each parameter increase, number of experiments will be increased dramatically.

Compare with DOE, GPRBOA method can significantly reduce number of experiments, but it's efficiency of find optimal parameters is relatively low, sometimes in less number of experiments can be to find the optimal parameters, and sometimes need much more experimental parameters to find the optimal parameters, This is due to uncertainty caused by random exploration. On the other hand, the solution found by GPRBOA may be not optimal, this is due to the use of change of hyperparameters as the convergence criterion, the assembly model may be premature convergence which means not convergence to the real assembly model, the cycle time of all parameter combinations can't be estimate correctly, therefore, such a solution by GPRBOA is not robust, when a change in the model, the obtained solution will become invalid, while the other solutions and not correct, only to relearn the model and re-optimization, this reduce low efficiency again.

The OE-GPRBOA based on orthogonal exploration, can comprehensive and efficient explore the parameter space, the convergence criterion used by average model learning deviation $avg\sigma$ can be better judge whether the model converges to the true model. Combination of orthogonal exploration and average model learning deviation $avg\sigma$ convergence criterion can minimize the number of experiments, and find the optimal parameters successfully.

V. CONCLUSION

In this paper, an online robotic assembly parameter optimization method OE-GPRBOA is proposed, which can make the industrial robot online find the optimal assembly parameters for precision assembly tasks, and complete assembly tasks in the shortest time. This method solves the problems of GPRBOA method about low efficiency, low success rate and low robustness. The OE-GPRBOA method use the theory of the orthogonal experimental design, Gaussian process regression and Bayesian Optimization, it has advantages such as high efficiency of exploration, effective modelling of assembly processes, good prediction and optimization for optimal assembly parameters. Experimental results show that OE-GPRBOA is more effective than DOE and GPRBOA, which can greatly improve the efficiency and success rate to find the optimal parameters. The online assembly parameters optimization method can be used for various assembly tasks.

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